BAREC-ST-2026: The Second Shared Task on Arabic Readability Assessment

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Abstract

We present the results and findings of the BAREC Shared Task 2026, the second shared task on Arabic readability assessment, organized as part of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026). The shared task focuses on automatic readability assessment using the BAREC Corpus (Elmadani et al., 2025b), addressing fine-grained classification across 19 readability levels. It comprises two subtasks—sentence-level classification and document-level classification—and three tracks: (1) the *Strict Track*, which permits only the BAREC Corpus; (2) the *Constrained Track*, which permits the BAREC Corpus, the SAMER Corpus (Alhafni et al., 2024), and the SAMER Lexicon (Al Khalil et al., 2020); and (3) the *Open Track*, which permits the use of any external resources. A total of 14 teams from 12 countries registered for the task, of which 10 submitted system description papers. All participating teams competed only in the sentence-level classification subtask. The top-performing system achieved a quadratic weighted kappa (QWK) of 85.3.[1]

RL	Grade	Example	
1	KG	Sara	سارة
3	1st	An egg in the nest!	بيضة في العش!
7	2nd		هل جئت بصيد وفير؟
		Did you come back with a good catch?	
10	4th		كان يجري مفزوعًا وكأنه يسابق الريح!
		He was running in fright, as though racing the wind!	
14	8th		الأبعاد الجيوسياسية لندرة المياه
		The geopolitical dimensions of water scarcity	
17	Uni		حيا وقال درتان في وعاء ممتهن
		He greeted [them] and said, "Two pearls in a humble vessel."	

Table 1: Examples by Reading Level (RL) and grade.

1 Introduction

Readability assessment plays an important role in education, literacy development, and language learning by helping ensure that texts are appropriate for readers' proficiency levels. Texts that are poorly matched to readers' abilities may adversely affect comprehension, retention, reading speed, and engagement (DuBay, 2004; Klare, 1963). Consequently, text-leveling systems are widely used, particularly in early education, to support structured and measurable progress in reading development (Allington et al., 2015; Barber and Klauda, 2020).

Although readability models have been developed for several languages, substantial challenges remain, especially for fine-grained text leveling and resource-scarce languages. For example, Fountas and Pinnell's 27-level framework for English (Fountas and Pinnell, 2006) and Taha-Thomure's (2017) 19-level framework for Arabic illustrate the value of detailed readability classifications. Such fine-grained levels enable more precise educational applications while remaining flexible enough to be mapped to broader categories when needed.

The Taha/Arabi21 framework (Taha-Thomure, 2017), which has been used to annotate more than 9,000 children's books, forms the basis of our work. Building on this framework, the BAREC guidelines (Habash et al., 2025) provide a standardized approach to sentence-level readability assessment across a broad range of genres and educational stages, from early childhood through postgraduate education(see Table 1). For the complete guidelines in Arabic and English, we refer readers to Habash et al. (2025).

Arabic presents distinctive challenges for readability assessment because of its rich morphology, extensive lexicon, and highly ambiguous orthography. In contrast to English, for which established readability formulas and datasets are available, Ara-

[1]https://barec.camel-lab.com/sharedtask2026

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bic readability research has long lacked standardized resources. This gap has limited the development of robust computational models that can accurately assess text difficulty across proficiency levels.

The BAREC Shared Task 2026 was organized as part of the Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026), co-located with EMNLP 2026. Fourteen teams from 12 countries registered for the shared task, and 10 teams submitted system description papers, which are cited throughout this overview (see §5). This paper describes the task design, summarizes the submitted systems, and presents the official results.

The paper is structured as follows: §2 reviews related work. §3 outlines the sub-tasks and tracks of the shared task. §4 introduces the datasets and evaluation metrics. §5 describes the baselines and provides an overview of the submitted systems. Finally, §6 reports and discusses the results.

2 Related Work

Automatic Readability Assessment Automatic readability assessment has been studied extensively across languages, yielding a wide range of datasets, features, and modeling approaches (Collins-Thompson and Callan, 2004; Pitler and Nenkova, 2008; Feng et al., 2010; Vajjala and Meurers, 2012; Xu et al., 2015; Nadeem and Ostendorf, 2018; Vajjala and Lučić, 2018; Deutsch et al., 2020; Lee et al., 2021). Early work, particularly for English, often relied on textbook collections, whose graded organization naturally supports readability evaluation (Vajjala, 2022). More recent work has expanded to alternative sources, including crowdsourced readability annotations from online platforms (Vajjala and Meurers, 2012; Vajjala and Lučić, 2018) and proficiency examinations designed for second-language learners under the CEFR framework (Xia et al., 2016). These efforts have also motivated increasingly fine-grained approaches to readability prediction, supporting more precise educational applications.

Arabic Readability Efforts Arabic readability research has evolved from handcrafted formulas based on surface features, such as word and sentence length (Al-Dawsari, 2004; Al-Khalifa and Al-Ajlan, 2010; Hazim et al., 2022), to machine-learning approaches that incorporate linguistic features, including part-of-speech tags, morphological information, diacritization, and script-specific characteristics (Forsyth, 2014; Saddiki et al., 2018; El-Haj and Rayson, 2016). Recent work has further demonstrated the effectiveness of pretrained language models for Arabic readability assessment (Liberato et al., 2024). Several graded resources have also been developed. The Taha/Arabi21 framework (Taha-Thomure, 2017) defines 19 readability levels for children's books and has been used to level more than 9,000 books. Other efforts include CEFR-based resources and vocabulary profiling (Habash and Palfreyman, 2022; Naous et al., 2024; Soliman and Familiar, 2024), the DARES datasets derived from Saudi school materials (El-Haj et al., 2024; Almujaiwel et al., 2025), and the SAMER project, which introduced a five-level readability lexicon (Al Khalil et al., 2020) and a manually annotated Arabic text simplification corpus (Alhafni et al., 2024).

BAREC Project This shared task is part of the broader BAREC project, which develops resources and tools for fine-grained Arabic readability assessment across the Arab world. The project is grounded in the BAREC Annotation Framework, which provides 19-level guidelines for Arabic readability annotation (Habash et al., 2025). Building on these guidelines, the BAREC Corpus provides a large-scale, manually annotated dataset for sentence-level readability assessment (Elmadani et al., 2025b), while BAREC Demo offers a web-based interface for assessing the readability of Arabic sentences (Altarbouch et al., 2025). The first BAREC Shared Task (Elmadani et al., 2025a) established a community benchmark for Arabic readability assessment, and the BAREC-10M Corpus subsequently extended the available data with a substantially larger automatically labeled corpus (Elmadani et al., 2026). The BAREC Shared Task 2026 builds on this foundation by providing a new evaluation setting for fine-grained sentence- and document-level Arabic readability classification.

History of BAREC Shared Task This is the second edition of the BAREC Shared Task. The first edition, BAREC Shared Task 2025, was organized as part of the Third Arabic Natural Language Processing Conference (ArabicNLP 2025), co-located with EMNLP 2025. It attracted 22 registered teams from 12 countries, 17 of which submitted system description papers. Most participating teams relied on pretrained language models (PLMs), often augmented with morphological features; several systems additionally used ensembling, label bal-

Split	#Documents	#Sentences	#Words
Train	1,518 (79%)	54,845 (79%)	832,743 (80%)
Dev	194 (10%)	7,310 (11%)	101,364 (10%)
Test	210 (11%)	7,286 (10%)	105,264 (10%)
All	1,922 (100%)	69,441 (100%)	1,039,371 (100%)
ST-2025	100	3,420	53,052
ST-2026	**212**	**8,077**	**110,981**

Table 2: **BAREC** Corpus splits.

ancing, or ordinal-aware loss functions. Teams such as !MSA (Basem et al., 2025), AMAR (Saeed et al., 2025), and Qais (Ahmed, 2025) also explored large language models (LLMs), whereas GNNinjas (Elchafei et al., 2025) and Syntaxa (Bahloul, 2025) investigated graph neural networks (GNNs) (Zhou et al., 2021), and Pixel (Sapirstein, 2025) explored vision-language models (Dosovitskiy et al., 2021; Rust et al., 2023). Traditional non-neural approaches were comparatively rare. Overall, the most successful systems combined PLMs with linguistic features and ensemble-based methods, establishing a strong baseline for subsequent BAREC shared-task editions.

The test set used in this edition is approximately twice the size of that used in the previous shared task (see Table 2). To better assess system generalization, it includes data from 17 publishing houses that were not represented in the original BAREC Corpus, enabling a more robust evaluation of a system's ability to handle previously unseen text sources.

3 Task Description

The BAREC Readability Assessment Shared Task evaluates systems for fine-grained Arabic readability classification under a 19-level framework. Participants were invited to develop models that predict readability at both the sentence and document levels.

Sub-tasks The shared task comprises the following two subtasks:

1. **Sentence-Level Classification:** Given an individual sentence, predict its readability level.
2. **Document-Level Classification:** Given a document consisting of consecutive sentences, predict its readability level. The document label is defined by the readability level of its most difficult sentence.

Tracks Each sub-task is offered under three tracks with different constraints on the resources that participants may use:

- **Strict Track:** Systems may be trained exclusively on the BAREC Corpus (Elmadani et al., 2025b).
- **Constrained Track:** Systems may use the BAREC Corpus, the SAMER Corpus (Alhafni et al., 2024) and the SAMER Lexicon (Al Khalil et al., 2020).
- **Open Track:** Systems may use any publicly available external resources.

Together, the two sub-tasks and three tracks define six possible subtask–track combinations. Participants were permitted to submit to multiple combinations.

4 Shared Task Datasets and Evaluation

This section describes the datasets available under the different tracks, the evaluation metrics, and the submission guidelines provided to participants.

4.1 Datasets

BAREC Corpus The BAREC Corpus (Elmadani et al., 2025b) is the primary dataset used in the shared task. It contains 1,922 documents and 69,441 sentences annotated with 19 readability levels. The corpus is partitioned at the document level into **Train** (≃80%), **Dev** (≃10%), and **Test** (≃10%) splits. Table 2 presents statistics for these splits at the document, sentence, and word levels.

SAMER Corpus The SAMER Corpus (Alhafni et al., 2024) contains 4,289 documents (158K words) and 20,358 fragments annotated with three readability levels. For this shared task, we use the fragment data made available and reported by Liberato et al. (2024). Table 4 summarizes the corpus statistics for the Train, Dev, and Test splits.

SAMER Lexicon The SAMER Lexicon (Al Khalil et al., 2020) is a readability lexicon for Modern Standard Arabic (MSA) containing 40K lemma–part-of-speech pairs annotated with five readability levels. The lexicon was manually annotated by three language professionals from different regions of the Arab world. Table 5 presents the distribution of readability levels in the lexicon.

				BAREC Corpus v1 (Sentences)											
Level-3	**Level-5**	**Level-7**	**Level-19**	**All**		**Train**		**Dev**		**Test**		**ST-2025**		**ST-2026**	
1	1	1	**1-alif**	409	1%	333	1%	44	1%	32	0%	21	1%	37	0%
			2-ba	437	1%	333	1%	68	1%	36	0%	21	1%	98	1%
			3-jim	1,462	2%	1,139	2%	182	2%	141	2%	69	2%	194	2%
			4-dal	751	1%	587	1%	78	1%	86	1%	28	1%	61	1%
		2	**5-ha**	3,443	5%	2,646	5%	417	6%	380	5%	188	5%	325	4%
			6-waw	1,534	2%	1,206	2%	189	3%	139	2%	47	1%	185	2%
			7-zay	5,438	8%	4,152	8%	701	10%	585	8%	296	9%	647	8%
	2	3	**8-Ha**	5,683	8%	4,529	8%	613	8%	541	7%	263	8%	775	10%
			9-ta	2,023	3%	1,597	3%	236	3%	190	3%	101	3%	310	4%
		4	**10-ya**	9,763	14%	7,741	14%	1,012	14%	1,010	14%	457	13%	1,547	19%
			11-kaf	4,914	7%	4,041	7%	409	6%	464	6%	233	7%	479	6%
2	3	5	**12-lam**	14,471	21%	11,318	21%	1,491	20%	1,662	23%	682	20%	1,713	21%
			13-mim	4,039	6%	3,252	6%	349	5%	438	6%	177	5%	405	5%
3	4	6	**14-nun**	10,687	15%	8,573	16%	1,072	15%	1,042	14%	596	17%	1,048	13%
			15-sin	2,547	4%	2,016	4%	258	4%	273	4%	171	5%	174	2%
	5	7	**16-ayn**	1,141	2%	866	2%	114	2%	161	2%	55	2%	49	1%
			17-fa	480	1%	364	1%	49	1%	67	1%	15	0%	23	0%
			18-sad	103	0%	67	0%	13	0%	23	0%	0	0%	0	0%
			19-qaf	116	0%	85	0%	15	0%	16	0%	0	0%	7	0%
Total				69,441	100%	54,845	100%	7,310	100%	7,286	100%	3,420	100%	8,077	100%

Table 3: Sentence-level splits across BAREC readability levels.

Split	**#Documents**	**#Fragments**	**#Words**
Train	2,790 (65%)	14,256 (70%)	112,828 (71%)
Dev	607 (14%)	2,948 (14%)	22,075 (14%)
Test	892 (21%)	3,154 (15%)	23,161 (15%)
All	4,289 (100%)	20,358 (100%)	158,064 (100%)

Table 4: SAMER Corpus splits.

Level	**Type Count**
Level I	3,545 (9%)
Level II	3,221 (8%)
Level III	5,510 (14%)
Level IV	10,130 (25%)
Level V	18,281 (45%)
Total	40,687 (100%)

Table 5: SAMER Lexicon distributions

BAREC-ST-2026 Blind Test Set As in the BAREC Shared Task 2025, we provide a new blind test set for the final evaluation, annotated according to the 19-level BAREC framework. This test set is approximately twice the size of that used in the previous edition. To better assess system generalization, it includes texts from 17 publishing houses that were not represented in the original BAREC Corpus. This design enables a more robust evaluation of systems' ability to handle text from previously unseen sources.

4.2 Evaluation Metrics

Following Elmadani et al. (2025b), we formulate readability assessment as an ordinal classification problem and evaluate submitted systems using the following metrics:

- **Accuracy (Acc):** The proportion of predictions that exactly match the reference labels under the 19-level scheme (**Acc**^{19}). We additionally report coarse-grained accuracy measures (**Acc**^{7}, **Acc**^{5}, and **Acc**^{3}) obtained by collapsing the 19 levels into 7, 5, and 3 levels, respectively (see Table 3).
- **Adjacent Accuracy ($\pm$1 Acc^{19}):** Also known as off-by-one accuracy, this metric considers a prediction correct when it exactly matches the reference label or differs from it by one readability level.
- **Average Distance (Dist):** Equivalent to mean absolute error (MAE), this metric measures the average absolute difference between predicted and reference labels.
- **Quadratic Weighted Kappa (QWK):** An extension of Cohen's kappa (Cohen, 1968; Doewes et al., 2023), QWK measures agree-

ment between predicted and reference labels while assigning quadratically greater penalties to errors that are farther from the reference label.

As in the first BAREC Shared Task, we use **QWK** as the primary metric for ranking systems. We prioritize QWK because it captures the ordinal structure of readability levels and provides a smooth, distance-sensitive penalty for misclassifications, unlike the hard decision boundaries used by accuracy-based measures.

4.3 Submission Guidelines

The shared task was organized in two phases: a development phase and a testing phase. During the **development phase**, we hosted CodaBench (Xu et al., 2022) competitions for all tracks. Participants could evaluate their systems locally using the BAREC development and test sets,[2] or submit predictions for the BAREC Test set through the corresponding CodaBench competition. Because the BAREC Test set is publicly available, this phase was open to all interested participants.

During the **testing phase**, the official blind test set was released exclusively to registered participants. Registered teams were required to submit a system description paper.

5 Participants and Systems

Table 6 summarizes the main modeling approaches adopted by the participating teams. All submitted systems fine-tuned pretrained language models, often combining them with ordinal-aware loss functions, custom decision thresholds, ensemble methods, and linguistic or surface-level features. A general trend across submissions is that systems exploring and integrating a broader range of complementary techniques tended to achieve stronger performance. In this section, we provide a brief description of each team's approach, followed by several baselines.

Thylinao (Silchenko, 2026) Thylinao participated in the sentence-level Strict and Open tracks, submitting the same system to both. They used an equal-weight ensemble of 14 fine-tuned encoder models spanning several pretrained Arabic and multilingual language models, loss functions, and input representations, including raw text, word-level input, and D3Tok morphological segmentation. The member scores were standardized before averaging, and the final predictions were produced using custom cutpoints placed based on validation-optimized thresholds. They achieved a QWK score of 85.3 and ranked first in both tracks.

ZHR (Zaher, 2026) ZHR participated in the sentence-level Strict and Open tracks. Their systems combined several pretrained Arabic encoders fine-tuned using five ordinal objectives, with CAMeL-Tools morphological segmentation (Obeid et al., 2020), greedy ensemble selection, and QWK-optimized decision thresholds. For the Strict Track, the team used target-prior calibration to account for label-distribution shift in the development set, achieving a QWK score of 85.0. For the Open Track, they additionally pretrained selected models on silver-labeled data from BAREC-10M (Elmadani et al., 2026), achieving a QWK score of 85.2.

Nile Nexus (Zaman and Rishta, 2026) Nile Nexus participated in the sentence-level Strict Track. They fine-tuned several Arabic pretrained language models using different ordinal-aware objectives, including soft ordinal cross-entropy, squared Earth Mover's Distance (L. Hou, 2017), direct ordinal regression, and CORAL-style cumulative-threshold loss (Cao et al., 2020; Shi et al., 2023). They combined the strongest models using validation-based greedy ensemble selection optimized for QWK and applied learned, level-specific decision thresholds to convert continuous scores into readability labels. Their system achieved a QWK score of 84.8, ranking third in the Strict Track.

Skywalker (Liang and Li, 2026) Skywalker participated in the sentence-level Strict Track. They fine-tuned a diverse set of Arabic pretrained language models using classification, regression, Earth Mover's Distance (L. Hou, 2017), and multitask objectives. They combined model scores, confidence estimates, rank-based features, and metadata through cross-fitted LightGBM components (Ke et al., 2017). The resulting continuous scores were fused and mapped to the 19 readability levels using validation-optimized monotonic thresholds. Their system achieved a QWK score of 84.2, ranking fourth in the Strict Track.

Viper (Khang et al., 2026) Viper participated in the sentence-level Strict Track. They trained

[2] https://github.com/CAMeL-Lab/barec-shared-task-2025

Team	Affiliation	Sentence-Level Tracks	QWK	Ling.	PLM -FT	Ord. Loss	Cust. Thr.	Ens.
Thylinao	NUS, Singapore	Strict, Open	85.3	✓	✓	✓	✓	✓
ZHR	Nile University, Egypt	Strict, Open	85.2	✓	✓	✓	✓	✓
Nile Nexus	CUET, Bangladesh	Strict	84.8		✓	✓	✓	✓
Skywalker	MBZUAI, UAE	Strict	84.2		✓	✓	✓	✓
Viper	UIT, VNU-HCM, Vietnam	Strict	84.1	✓	✓	✓		✓
HP	UIT, VNU-HCM, Vietnam	Strict	83.7		✓	✓	✓	✓
NAMAA	AIU, Syria	Strict	82.4	✓	✓	✓		✓
LSI	LSI, USA	Constrained, Open	81.7		✓			✓
Soloeval	Habib University, Pakistan	Open	80.6	✓	✓	✓		
Northwestern_2	NU-Q, Qatar	Strict	79.0		✓			

Table 6: Summary of the participating teams, their affiliations, sentence-level tracks, best official QWK scores, and reported modeling approaches. QWK scores are reported as percentages; for teams participating in multiple tracks, the highest score is shown. *Ling.* indicates the use of linguistic or surface-level features, *PLM-FT* the fine-tuning of pretrained language models, *Ord. Loss* an ordinal-aware training objective, *Cust. Thr.* custom decision-threshold optimization, and *Ens.* model ensembling. A checkmark denotes that the corresponding technique was used.

Team	Track	QWK	Acc^{19}	$\pm1\ \text{Acc}^{19}$	Dist	Acc^{7}	Acc^{5}	Acc^{3}
Thylinao[†]	Strict	**85.3**	37.7	70.8	1.14	60.5	66.9	74.6
ZHR	Open	85.2	39.6	72.1	1.08	61.4	68.5	75.8
ZHR	Strict	85.0	38.8	72.0	1.10	60.4	68.7	75.8
Nile Nexus	Strict	84.8	37.9	73.5	1.10	61.4	68.4	75.1
Skywalker	Strict	84.2	41.7	73.3	1.10	60.9	69.2	76.1
Viper	Strict	84.1	40.9	**73.6**	1.10	62.3	69.2	75.7
HP	Strict	83.7	47.7	72.6	1.05	63.5	69.6	76.4
Baseline III	Strict	82.5	42.4	**73.6**	1.09	61.6	68.8	75.6
NAMAA	Strict	82.4	37.1	73.2	1.10	59.5	67.2	74.7
LSI[‡]	Constrained	81.7	50.6	72.9	**1.00**	65.0	71.2	77.2
Baseline II	Strict	81.1	**59.2**	71.8	1.01	**68.3**	**72.5**	**77.3**
soloeval[§]	Strict	80.6	49.1	**73.6**	1.10	64.5	70.0	75.1
Baseline I	Strict	79.7	58.2	71.1	1.04	66.6	72.4	77.0
Northwestern_2	Strict	79.0	55.6	70.0	1.10	65.3	70.5	76.2

Table 7: Official sentence-level results for the participating teams and organizer baselines, ordered by QWK. [†]The Open Track result is omitted because the same Thylinao system was submitted to both the Strict and Open tracks. [‡]The Open Track result is omitted because the same LSI system was submitted to both the Constrained and Open tracks. [§]Although Soloeval submitted to the Open Track, their system is listed as Strict-equivalent because it used only resources permitted under the Strict Track. All scores except Dist are reported as percentages.

five AraBERTv2-based (Antoun et al., 2020) regression models using paired representations that combined D3Tok morphological segmentation with normalized surface text, sentence-level statistics, and corpus metadata. To mitigate class imbalance, they applied level-aware weighted sampling during training. The five models were trained with different random seeds and combined through uniform averaging, after which the continuous ensemble score was rounded to the nearest readability level. Their system achieved a QWK score of 84.1, ranking fifth in the Strict Track.

HP (Dinh and Van, 2026) HP participated in the sentence-level Strict Track. They applied parameter-efficient fine-tuning with LoRA (Hu et al., 2022) to AraBERTv02 (Antoun et al., 2020) and ARBERT (Abdul-Mageed et al., 2021), formulating the task as ordinal regression with a sample-weighted CORAL loss. They trained their models using stratified five-fold cross-validation and multiple random seeds. For post-processing, they combined predictions obtained from QWK-optimized Nelder–Mead thresholds with distribution-aligned predictions through an equal-weight blend. Their best system, based on AraBERTv02, achieved a QWK score of 83.7 on the blind test set.

NAMAA (Sibaee et al., 2026) NAMAA participated in the sentence-level Strict Track. They evaluated a progression of approaches ranging from engineered linguistic and morphological features and TF–IDF representations to contextual transformer models, multi-encoder fusion, and

output-level ensembling. Their final submission used EmbeddingGemma-300M (Schechter Vera et al., 2025) with Gaussian soft labels, an ordinal KL-divergence objective, inverse-frequency class weighting, concatenated [CLS] and mean-pooled representations, and expected-value decoding. They selected the best checkpoint according to development-set QWK. Their system achieved a QWK score of 82.4 on the blind test set.

LSI (Al-Madi, 2026) LSI participated in the sentence-level Constrained and Open tracks, submitting the same system to both. They built an equal-weight, logit-level ensemble of three complementary Arabic pretrained language models: AraBERT (Antoun et al., 2020), AraModernBERT(Elshehy et al., 2026), and CAMeLBERT-Mix (Inoue et al., 2021). AraBERT and AraModernBERT were trained with cross-entropy, whereas CAMeLBERT-Mix used an ordinal Earth Mover's Distance loss and incorporated six sentence-level lexical-difficulty features derived from the SAMER Lexicon. Their ensemble achieved a QWK score of 81.7, ranking first in the Constrained Track and third in the Open Track.

Soloeval (Hussain Usmani, 2026) Soloeval participated in the sentence-level Open Track. They fine-tuned MARBERTv2 (Abdul-Mageed et al., 2021) and CAMeLBERT-Mix (Inoue et al., 2021) separately using a soft-label ordinal classification objective, in which Gaussian target distributions centered on the gold levels encouraged distance-sensitive predictions aligned with QWK. They augmented the encoder representations with five surface features capturing sentence length, word count, average word length, diacritic density, and punctuation density. Their final submission used CAMeLBERT-Mix with three-fold cross-validation and averaged the predictions of the fold-specific models. Although the Open Track permitted additional resources, they used only the official BAREC data and achieved a QWK score of 80.6.

Northwestern_2 (Shlkamy and Zaghouani, 2026) Northwestern_2 participated in the sentence-level Strict Track. They formulated readability assessment as standard 19-class classification and fine-tuned CAMeLBERT-Mix using cross-entropy loss. Sentences were tokenized with the model's native tokenizer, truncated to 128 subword tokens, and dynamically padded during training. The team selected the best checkpoint according to development-set QWK, and their submitted system achieved a QWK score of 79.0 on the blind test set.

Baselines We evaluated three baselines for comparison with the participating systems. All three are Arabic-specific BERT-base models fine-tuned on the BAREC Corpus and selected from the models introduced by Elmadani et al. (2025b). The baselines differ along three dimensions:

- **Pretrained language model:** AraBERTv02 or AraBERTv2 (Antoun et al., 2020).
- **Input representation: Word**, which applies basic sentence preprocessing, including the removal of diacritics and kashida; or **D3Tok**, which uses morphological segmentation to separate words into their base and clitic components.[3]
- **Training objective:** cross-entropy classification (**CE**) or regression with mean squared error (**Reg**).

Based on these dimensions, we selected the following baselines:

- **AraBERTv02+Word+CE (Baseline I):** A standard classification baseline that combines the widely used AraBERTv02 model with word-level input and a cross-entropy objective.
- **AraBERTv2+D3Tok+CE (Baseline II):** A classification baseline designed to measure the effect of linguistically motivated D3Tok segmentation.
- **AraBERTv2+D3Tok+Reg (Baseline III):** A regression baseline motivated by the ordered structure of the readability labels, for which the distance between the predicted and reference levels is meaningful.

6 Results

Table 7 presents the official results on the sentence-level blind test set, together with the baselines. No teams submitted systems for the document-level subtask. For teams that submitted an identical system to multiple tracks, duplicate results are omitted as described in the table caption.

By linking the results in tables 6 and 7, we see that the strongest systems generally combined several complementary techniques. In addition to fine-

[3]The D3Tok representation is generated using CAMeL Tools (Obeid et al., 2020).

tuning pretrained language models, the highest-ranked approaches typically incorporated ordinal-aware objectives, customized decision thresholds, and model ensembling. Several systems also exploited morphological, linguistic, or surface-level information. This pattern suggests that combining diverse modeling and calibration strategies generally led to better performance. Access to additional resources was also not consistently decisive: the improvement obtained by the leading Open Track system over its Strict counterpart was small.

The results further highlight the distinction between exact classification accuracy and ordinal agreement. Some systems achieved comparatively high exact or coarse-grained accuracy while receiving lower QWK scores, whereas the highest-ranked systems under QWK did not necessarily obtain the best exact accuracy. This indicates that correctly modeling the ordering of the labels and avoiding distant errors are more important for QWK than maximizing exact matches alone. The baseline comparison supports this observation: the regression-based baseline achieved the strongest baseline QWK, while a cross-entropy baseline performed better on several accuracy measures. Overall, these findings reinforce the importance of ordinal-aware training, ensemble diversity, and prediction calibration for fine-grained Arabic readability assessment.

7 Conclusion

We organized the BAREC Shared Task 2026, the second shared task on fine-grained Arabic readability assessment. The task offered sentence- and document-level classification under Strict, Constrained, and Open resource settings, using the 19-level BAREC framework. 14 teams from 12 countries registered, and 10 submitted system description papers. The official blind test set included texts from 17 previously unseen publishing houses, provided a challenging evaluation of cross-source generalization. The strongest systems combined pretrained language models with ordinal-aware objectives, custom thresholds, linguistic features, and ensembling. We hope this shared task encourages further research on robust sentence- and document-level Arabic readability assessment.

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Limitations

This work has several limitations. First, The observed relationship between broader use of ordinal objectives, custom thresholds, linguistic features, and ensembling and stronger performance is correlational: the submitted systems differ in many respects, preventing the contribution of individual components from being isolated. Second, we use Quadratic Weighted Kappa (QWK) as the primary evaluation metric because it accounts for the ordinal nature of the readability levels. Nevertheless, determining the most appropriate metric for fine-grained readability assessment remains an open research question.

Ethics Statement

We used AI writing assistance within the scope of "Assistance purely with the language of the paper" described in the ACL Policy on Publication Ethics.

References

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Samar Ahmed. 2025. Qais at BAREC Shared Task 2025: A Fine-Grained Approach for Arabic Readability Classification Using a pre-trained model. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

M Al-Dawsari. 2004. The assessment of readability books content (boys-girls) of the first grade of intermediate school according to readability standards. *Sultan Qaboos University, Muscat*.

Hend S Al-Khalifa and Amani A Al-Ajlan. 2010. Automatic readability measurements of the Arabic text: An exploratory study. *Arabian Journal for Science and Engineering*, 35(2 C):103–124.

Muhamed Al Khalil, Nizar Habash, and Zhengyang Jiang. 2020. A large-scale leveled readability lexicon for Standard Arabic. In *Proceedings of the*

Twelfth Language Resources and Evaluation Conference, pages 3053–3062, Marseille, France. European Language Resources Association.

Sahara H. Al-Madi. 2026. LSI at BAREC Shared Task 2026: A Controlled Study of Training Convergence, Heuristics, and Ensembles for Arabic Sentence-Level Readability. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Bashar Alhafni, Reem Hazim, Juan David Pineros Liberato, Muhamed Al Khalil, and Nizar Habash. 2024. The SAMER Arabic text simplification corpus. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16079–16093, Torino, Italia. ELRA and ICCL.

Richard L Allington, Kimberly McCuiston, and Monica Billen. 2015. What research says about text complexity and learning to read. *The Reading Teacher*, 68(7):491–501.

Sultan Almujaiwel, Damith Premasiri, Tharindu Ranasinghe, Mo El-Haj, and Ruslan Mitkov. 2025. Complex concept-based readability estimation from arabic curriculum. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.*, 24(11).

Kinda Altarbouch, Khalid N. Elmadani, Ossama Obeid, Hanada Taha, and Nizar Habash. 2025. BAREC demo: Resources and tools for sentence-level Arabic readability assessment. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 950–959, Suzhou, China. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.

Ahmed Bahloul. 2025. Syntaxa at BAREC Shared Task 2025: BERTnParse - Fusion of BERT and Dependency Graphs for Readability Prediction. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

Amelia T. Barber and Susan L. Klauda. 2020. How reading motivation and engagement enable reading achievement: Policy implications. *Policy Insights from the Behavioral and Brain Sciences*, 7(1):27–34.

Mohamed Basem, Mohammed Younes, Seif Ahmed, and Abdelrahman Moustafa. 2025. !MSA at BAREC Shared Task 2025: Ensembling Arabic Transformers for Readability Assessment. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. 2020. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331.

Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement provision for scaled disagreement or partial credit. *Psychological bulletin*, 70(4):213.

Kevyn Collins-Thompson and James P. Callan. 2004. A language modeling approach to predicting reading difficulty. In *Proceedings of the Human Language Technology Conference of the North American Chapter of the Association for Computational Linguistics: HLT-NAACL 2004*, pages 193–200, Boston, Massachusetts, USA. Association for Computational Linguistics.

Tovly Deutsch, Masoud Jasbi, and Stuart Shieber. 2020. Linguistic features for readability assessment. In *Proceedings of the Fifteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 1–17, Seattle, WA, USA → Online. Association for Computational Linguistics.

Hoang Phong Dinh and Thin Dang Van. 2026. HP at BAREC Shared Task 2026: Parameter-Efficient Ordinal Regression and Threshold Optimization for Arabic Readability. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Afrizal Doewes, Nughthoh Arfawi Kurdhi, and Akrati Saxena. 2023. Evaluating quadratic weighted kappa as the standard performance metric for automated essay scoring. In *Proceedings of the 16th International Conference on Educational Data Mining*, pages 103–113, Bengaluru, India. International Educational Data Mining Society.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An image is worth 16x16 words: Transformers for image recognition at scale. *Preprint*, arXiv:2010.11929.

William H DuBay. 2004. The principles of readability. *Online Submission*.

Mahmoud El-Haj and Paul Rayson. 2016. OSMAN — a novel Arabic readability metric. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, pages 250–255, Portorož, Slovenia. European Language Resources Association (ELRA).

Mo El-Haj, Sultan Almujaiwel, Damith Premasiri, Tharindu Ranasinghe, and Ruslan Mitkov. 2024. DARES: Dataset for Arabic readability estimation of school materials. In *Proceedings of the Workshop on DeTermIt! Evaluating Text Difficulty in a Multilingual Context @ LREC-COLING 2024*, pages 103–113, Torino, Italia. ELRA and ICCL.

Passant Elchafei, Mayar Osama, Mohamed Rageh, and Mervat Abu-Elkheir. 2025. GNNinjas at BAREC Shared Task 2025: Lexicon-Enriched Graph Modeling for Arabic Document Readability Prediction. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025a. BAREC shared task 2025

on Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.

Khalid N. Elmadani, Nizar Habash, and Hanada Taha-Thomure. 2025b. A large and balanced corpus for fine-grained Arabic readability assessment. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16376–16400, Vienna, Austria. Association for Computational Linguistics.

Khalid N. Elmadani, Adel Mahmoud Wizani, Hanada Taha Thomure, and Nizar Habash. 2026. A large and balanced multi-domain Arabic corpus annotated for morphology, syntax, and readability. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 11761–11775, Palma de Mallorca, Spain. ELRA Language Resource Association.

Omar Elshehy, Omer Nacar, Abdelbasset Djamai, Muhammed Ragab, Khloud Al Jallad, and Mona Abdelazim. 2026. AraModernBERT: Transtokenized initialization and long-context encoder modeling for Arabic. In *Proceedings of the 2nd Workshop on NLP for Languages Using Arabic Script*, pages 313–321, Rabat, Morocco. Association for Computational Linguistics.

Lijun Feng, Martin Jansche, Matt Huenerfauth, and Noémie Elhadad. 2010. A comparison of features for automatic readability assessment. In *Coling 2010: Posters*, pages 276–284, Beijing, China. Coling 2010 Organizing Committee.

Jonathan Forsyth. 2014. Automatic readability prediction for modern standard Arabic. In *Proceedings of the Workshop on Open-Source Arabic Corpora and Processing Tools (OSACT)*.

Irene C Fountas and Gay Su Pinnell. 2006. *Leveled books (k-8): Matching texts to readers for effective teaching*. Heinemann Educational Books.

Nizar Habash and David Palfreyman. 2022. ZAEBUC: An annotated Arabic-English bilingual writer corpus. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 79–88, Marseille, France. European Language Resources Association.

Nizar Habash, Hanada Taha-Thomure, Khalid N. Elmadani, Zeina Zeino, and Abdallah Abushmaes. 2025. Guidelines for fine-grained sentence-level Arabic readability annotation. In *Proceedings of the 19th Linguistic Annotation Workshop (LAW-XIX-2025)*, pages 359–376, Vienna, Austria. Association for Computational Linguistics.

Reem Hazim, Hind Saddiki, Bashar Alhafni, Muhamed Al Khalil, and Nizar Habash. 2022. Arabic word-level readability visualization for assisted text simplification. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 242–249, Abu Dhabi, UAE. Association for Computational Linguistics.

Edward J Hu, yelong shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.

Turab Hussain Usmani. 2026. Soloeval at BAREC Shared Task 2026: QWK-Aligned Soft-Label Ordinal Classification with Auxiliary Surface Features for Arabic Readability Assessment. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

Nguyen Minh Khang, Nguyen Ngoc Nho, Trong Thanh Le, and Thin Dang Van. 2026. Viper at BAREC Shared Task 2026: Structured Arabic Representations and Robust Multi-Seed Ensembling for Readability Assessment. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

G.R. Klare. 1963. *The Measurement of Readability*. Iowa State University Press.

D. Samaras L. Hou, C.P. Yu. 2017. Squared earth mover's distance loss for training deep neural networks on ordered-classes. In *NIPS workshop on Learning on Distributions, Functions, Graphs and Groups*.

Bruce W. Lee, Yoo Sung Jang, and Jason Lee. 2021. Pushing on text readability assessment: A transformer meets handcrafted linguistic features. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 10669–10686, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Junhong Liang and Xueqi Li. 2026. Skywalker at BAREC Shared Task 2026: Multi-Expert PLM Fusion and QWK-Aware Ordinal Decoding. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Juan Liberato, Bashar Alhafni, Muhamed Khalil, and Nizar Habash. 2024. Strategies for Arabic readability modeling. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 55–66, Bangkok, Thailand. Association for Computational Linguistics.

Farah Nadeem and Mari Ostendorf. 2018. Estimating linguistic complexity for science texts. In *Proceedings of the Thirteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 45–55, New Orleans, Louisiana. Association for Computational Linguistics.

Tarek Naous, Michael J Ryan, Anton Lavrouk, Mohit Chandra, and Wei Xu. 2024. ReadMe++: Benchmarking multilingual language models for multi-domain readability assessment. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 12230–12266, Miami, Florida, USA. Association for Computational Linguistics.

Ossama Obeid, Nasser Zalmout, Salam Khalifa, Dima Taji, Mai Oudah, Bashar Alhafni, Go Inoue, Fadhl Eryani, Alexander Erdmann, and Nizar Habash. 2020. CAMeL tools: An open source python toolkit for Arabic natural language processing. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 7022–7032, Marseille, France.

Emily Pitler and Ani Nenkova. 2008. Revisiting readability: A unified framework for predicting text quality. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, pages 186–195, Honolulu, Hawaii. Association for Computational Linguistics.

Phillip Rust, Jonas F. Lotz, Emanuele Bugliarello, Elizabeth Salesky, Miryam de Lhoneux, and Desmond Elliott. 2023. Language modelling with pixels. *Preprint*, arXiv:2207.06991.

Hind Saddiki, Nizar Habash, Violetta Cavalli-Sforza, and Muhamed Al Khalil. 2018. Feature optimization for predicting readability of arabic l1 and l2. In *Proceedings of the 5th Workshop on Natural Language Processing Techniques for Educational Applications*, pages 20–29.

Mostafa Saeed, Rana Waly, and Abdelaziz Ashraf Hussein. 2025. AMAR at BAREC Shared Task 2025: Arabic Meta-learner for Assessing Readability. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

Ben Sapirstein. 2025. Pixels at BAREC Shared Task 2025: Visual Arabic Readability Assessment. In *Proceedings of the Third Arabic Natural Language Processing Conference*, Suzhou, China. Association for Computational Linguistics.

Henrique* Schechter Vera, Sahil* Dua, Biao Zhang, Daniel Salz, Ryan Mullins, Sindhu Raghuram Panyam, Sara Smoot, Iftekhar Naim, Joe Zou, Feiyang Chen, Daniel Cer, Alice Lisak, Min Choi, Lucas Gonzalez, Omar Sanseviero, Glenn Cameron, Ian Ballantyne, Kat Black, Kaifeng Chen, and 69 others. 2025. Embeddinggemma: Powerful and lightweight text representations.

Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Anal. Appl.*, 26(3):941–955.

Eman Sedqy Shlkamy and Wajdi Zaghouani. 2026. Northwestern 2 at BAREC Shared Task 2026: CAMeLBERT-Mix for Fine-Grained Arabic Readability Assessment. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Serry Sibaee, Khloud Al Jallad, Israa Elhosiny, Fatimah Emad Eldin, and Omer Nacar. 2026. NAMAA Community at BAREC 2026: Ordinal-Aware Training for Fine-Grained Arabic Readability Assessment Beyond Architectural Scale. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Maksim Silchenko. 2026. thylinao at BAREC Shared Task 2026: Ensembling at the Noise Floor. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Rasha Soliman and Laila Familiar. 2024. Creating a cefr arabic vocabulary profile: A frequency-based multi-dialectal approach. *Critical Multilingualism Studies*, 11(1):266–286.

Hanada Taha-Thomure. 2017. *Arabic Language Text Leveling* (معايير هنادا طه لتصنيف مستويات النصوص العربية). Educational Book House (دار الكتاب التربوي للنشر والتوزيع).

Sowmya Vajjala. 2022. Trends, limitations and open challenges in automatic readability assessment research. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 5366–5377, Marseille, France. European Language Resources Association.

Sowmya Vajjala and Ivana Lučić. 2018. OneStopEnglish corpus: A new corpus for automatic readability assessment and text simplification. In *Proceedings of the Thirteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 297–304, New Orleans, Louisiana. Association for Computational Linguistics.

Sowmya Vajjala and Detmar Meurers. 2012. On improving the accuracy of readability classification using insights from second language acquisition. In *Proceedings of the Seventh Workshop on Building Educational Applications Using NLP*, pages 163–173, Montréal, Canada. Association for Computational Linguistics.

Menglin Xia, Ekaterina Kochmar, and Ted Briscoe. 2016. Text readability assessment for second language learners. In *Proceedings of the 11th Workshop on Innovative Use of NLP for Building Educational Applications*, pages 12–22, San Diego, CA. Association for Computational Linguistics.

Wei Xu, Chris Callison-Burch, and Courtney Napoles. 2015. Problems in current text simplification research: New data can help. *Transactions of the Association for Computational Linguistics*, 3:283–297.

Zhen Xu, Sergio Escalera, Adrien Pavão, Magali Richard, Wei-Wei Tu, Quanming Yao, Huan Zhao, and Isabelle Guyon. 2022. Codabench: Flexible, easy-to-use, and reproducible meta-benchmark platform. *Patterns*, 3(7):100543.

Mahmoud Zaher. 2026. ZHR at BAREC Shared Task 2026: Ordinal-Objective Ensembles with Threshold Calibration and Silver-Label Pretraining for Arabic Readability Assessment. In *Proceedings of the*

Fourth Arabic Natural Language Processing Conference, Budapest, Hungary. Association for Computational Linguistics.

Sumaiya Zaman and Miftahul Jannat Rishta. 2026. Nile Nexus at BAREC Shared Task 2026: Enhancing Fine-Grained Arabic Readability Assessment through Ordinal-Aware Transformer Ensembling and Decision Calibration. In *Proceedings of the Fourth Arabic Natural Language Processing Conference*, Budapest, Hungary. Association for Computational Linguistics.

Jie Zhou, Ganqu Cui, Shengding Hu, Zhengyan Zhang, Cheng Yang, Zhiyuan Liu, Lifeng Wang, Changcheng Li, and Maosong Sun. 2021. Graph neural networks: A review of methods and applications. *Preprint*, arXiv:1812.08434.